

Facile preparation of multiscale graphene-basalt fiber reinforcements and their enhanced mechanical and tribological properties for polyamide 6 composites

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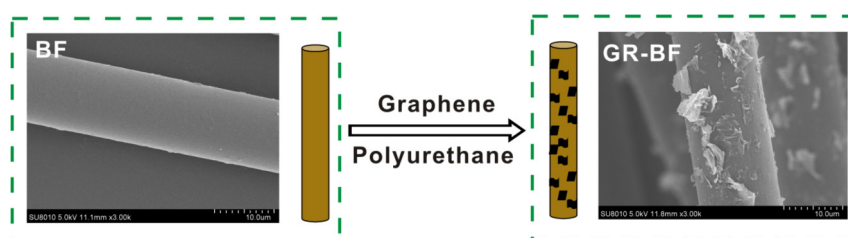
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HIGHLIGHTS

- Homogeneous graphene was facilely coated on the surface of basalt fibers.
- Interfacial adhesive can be improved for the basalt fibers reinforced composites.
- Multiscale structure enhanced the mechanical and tribological for the composites.
- Provided a facile and effective strategy to improve the performance of composites.

GRAPHICAL ABSTRACT



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ABSTRACT

To obtain strong interfacial adhesion between basalt fiber (BF) and polyamide 6 (PA6) matrix, a multiscale graphene-basalt fiber reinforcement was introduced by directly coating graphene (GR) onto the surface of individual basalt fiber via polyurethane sizing agent. The GR on the surface of BF was homogeneous, and the microstructure of the interfacial region between BF and matrix could be adjusted by varying the content of GR in the sizing agent. Owing to the bridge effect of GR, a mechanical meshing nanoscale interfacial area was generated, which could enhance the interfacial adhesive between BF and PA6 matrix, and the stress transfer of the corresponding composites was improved. The uniform coating of GR on the surface of BF increased the storage modulus and Tg for the BF/PA6 composites and 18.2% and 34% improvement for their tensile and flexural strength. Compared with BF/PA6, the GR-BF/PA6 composites obtained a fairly low friction coefficient and highly wear-resisting property, which was explained by the improved interfacial adhesion, effective stress transfer and the highly unfolded feature and flexibility of the introduced GR. This work presents a suitable approach for industrial production that can improve the comprehensive performance of fiber-reinforced composites effectively.

1. Introduction

With the increasing need of replacing metal counterparts in various

high performance structural applications in aerospace, automotive and mechanical industry, the investigation of basalt fibers (BF)-reinforced polymer composites (BFPCs) has been actively pursued owing to the

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excellent advantages of basalt fibers such as non-pollution, high strength, good thermal stability, no easy to damage, good friction stability at high temperature, low braking noise and good heat fading resistance. Comparison with glass fiber, asbestos fiber and other traditional fiber reinforced polymer composites, the BFPCs shows much higher performance and can be more likely to replace metal counterparts in many fields [1–3]. For instance, served as structural composite material applied in automobile brakes and clutches, the BFPCs not only boosts the high-speed development of the automobile, but also helps to improve the safety performance of the automobile. However, one of the shortcomings for most BFPCs is the weak interfacial bonding between fiber and polymer owing to the small specific surface area, low surface energy and high chemical inertness of basalt fiber. The weak interfacial bonding always leads to worse stress transfer and serious physical defects, which shortens the life of BFPCs and restricts their wide application to a large extent [4,5].

Modifying suitable polymeric sizing or coating on the surface of basalt fibers can improve the interfacial interaction between basalt fibers and polymer matrix to a certain extent. As an additional phase, the sizing layer can dominate stress transfer through the gradient physical properties thus drastically affects the final mechanical properties of composites. With the increase of surface roughness, the mechanical interlocking between fibers and matrix also can be improved by the sizing. Besides, the dissolution of sizing into the matrix will improve the chemical interaction between the sized fibers and matrix [6–9]. In order to gain a stronger interfacial interaction between polymer matrix and basalt fiber, BF has been surface treated by numerous approaches, such as electro-chemical oxidation [10–12], oxidation in strong acids [13–15], UV oxidation [16,17], chemical grafting [18–21], plasma treatment [22], and plasma polymerization [23]. However, these methods often cause damage to the surface and interior of fiber, which will be inevitable to affect the overall strength of the fiber. And these treating methods often need high-lever equipment, complicated process, long preparation period and high production cost. Therefore, it is of great interest to develop new concepts for improving the interface adhesive between basalt fiber and polymer matrix.

Recently, multiscale reinforcement, that introduce nanomaterials such as nanotubes or nanoparticles into the interface of fibers reinforced polymer composites, has been proved to be an efficient approach to achieve this aim [24,25]. With the characteristic of high activity and large specific surface area, the introduced nanomaterials will improve the interfacial adhesive between fibers and polymer matrix efficiently. And the performance of multi-scale enhancement materials can be improved synergistically by the combination of conventional micron fibers and nanofillers. Therefore, building multi-scale enhancement has attracted a wide spread attention to overcome the poor interface interaction for fibers reinforced polymer composites [26–28]. Two-dimensioning graphene (GR) possesses exhibits extraordinary mechanical, thermal properties, and the addition of graphenes with low content will improve the toughness of matrix and enhance the interfacial performance for the composites efficiently [29–34]. So building multiscale reinforcement using graphenes into the matrix or on the surface of one-dimensional fiber will be efficient to improve the interfacial properties for the basalt fiber reinforced composites.

In this study, we explored a facile method to prepare multiscale graphene-basalt fiber reinforcement via polyurethane sizing agent, and researched their enhancement on the mechanical and tribological properties for polyamide 6 composites. The graphene was dispersed in polyurethane sizing agent and then coated on the surface of basalt fiber directly. The morphology, dispersion and optimal concentration of GR on the surface of basalt fiber were discussed in detail. The enhanced mechanical and tribological properties of multiscale graphene-basalt fiber reinforced PA6 composites on account of their improved

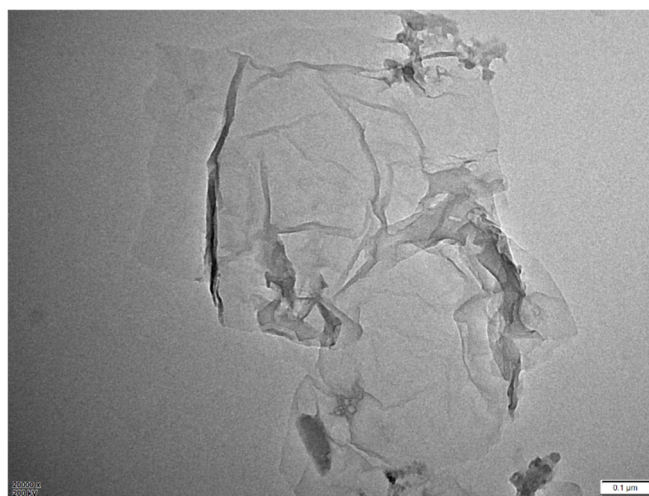


Fig. 1. TEM image of graphene.

interfacial effect were characterized and confirmed by SEM, tensile and flexural tests, DMA, friction and wear tests. This work offered a novel method to improve the interfacial, mechanical and tribological properties for basalt fiber filling composites by introducing multiscale structure into the interface.

2. Experimental details

2.1. Materials

Basalt fiber (BC13-2400W) with mean diameter of 13 μm was produced by Shanxi Jintou basalt Development Co. Ltd. polyamide 6 (PA6, 1013B) was obtained from Ube Industries (Japan). The graphene was obtained from Tangshan Jianhua Technology Development Co., Ltd (China). It was a black powder with the diameter 25–35 μm , bulk density about 5 mg/cm^3 , electrical conductivity 0.1–2 s/cm , specific surface area 400–550 m^2/g (BET method) and average number of layers about 6 layers. Fig. 1 showed the transmission electron microscope (TEM) image of the graphene obtained on a JEM-2100F TEM system. Water soluble polyurethane (MR-709) with the solid content of 40% was obtained from Dongguan Rice Chemical Co., Ltd (China).

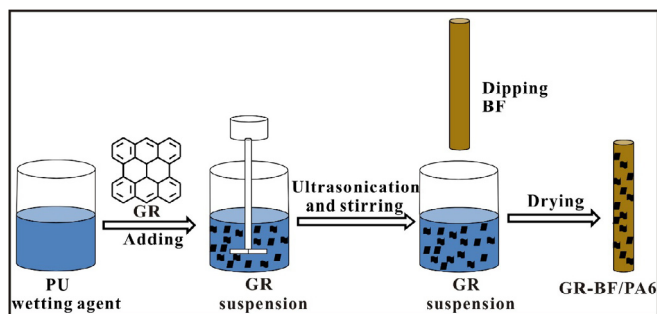
2.2. Preparation process

2.2.1. Preparation of graphene coated BF

Before being used, BF were put into acetone and treated by ultrasonic for 4 h to remove the commercial coating, which were denoted as BF. The polyurethane solution was diluted by deionized water to the concentration of 1 wt%. Then 0.1, 0.3, 0.5 and 1 g/L graphene were added into the polyurethane solution. Followed by ultrasonication and stirring for 20 min, a stable suspensions of graphene was obtained. Subsequently, 20 g BF was dipped into 1 L graphene suspensions for 10 min. Finally, the fibers were dried at 80 $^{\circ}\text{C}$ for 12 h. The BF treated by pure polyurethane solution was marked as PU-BF, and the BF coated by different concentration of GR were designated as 0.1 GR-BF, 0.3 GR-BF, 0.5 GR-BF and 1.0 GR-BF, respectively. (see Scheme 1).

2.2.2. Preparation of GR-BF/PA6 composites

The as-obtained 20 wt% GR-BF were mixed with 80 wt% PA6 to prepare GR-BF/PA6 composite samples using a TE-20 co-rotating twin-screw extruder (Nanjing Keya Coperion Machinery Co. Ltd., China) at a



Scheme 1. Coating process of graphene onto the surface of basalt fiber.

screw speed of 30 rpm and feed rate of 15 rpm. The temperature profile of the barrel was set at 225–230–235–240–230 °C from the hopper to the die. The materials were crushed and dried at 80 °C for 6 h. Then standard tensile and flexural test bars were prepared using SZ-100/80 plastic-injection molding machine (Shanghai Plastic Machinery Factory, China). The temperature of each stage of the injection molding machine was 225–235–240–245–250 °C.

2.3. Characterization methods

The scanning electron microscopy (SEM) images of the basalt fibers and fracture surface of the composite were obtained on a Hitachi SU-8010 field-emission SEM system (operated at 10 kV). The measurement of surface area of basalt fiber was carried out on the Autosorb-iQ automatic ratio surface and pore size distribution analyzer. The tensile and flexural properties of the PA6 composites were conducted on AI-7000s universal electronic testing machine according to ISO 178–2010 and ISO 527–2012. The dynamic mechanical analysis (DMA, Mettler-Toledo DMA/SDTA861e) were conducted in single cantilever mode between 30 and 160 °C at the frequency of 1 Hz and heating rate of 3 °C/min. The friction and wear tests were performed on a multifunctional material surface comprehensive performance tester (CFT-I, Lanzhou Zhongke Kaihua Technology Development Co., Ltd) under dry conditions. The dual samples was beared stainless ball with the diameter of 5 mm. The friction coefficient (μ) was exported automatically via the computer dealing with collected data and the wear volume loss (ΔV , mm³) was obtained according to real wear tracks measured by a micrometer (resolution: 0.0001 mm). The wear rate (K , mm³N^{−1} m^{−1}) of the specimen was calculated from the following equation:

$$k = \frac{\Delta V}{FtL} \quad (1)$$

Where F is the applied load (N), t is the experimental duration (min), and L is the stroke length in one cycle (m). In this work, three readings of the friction and wear tests of steady-state sliding under dry conditions were taken, and the average values were adopted in our results. The worn surfaces of the PA6 composites were examined by SEM and three dimensional non-contact surface topography instrument and profile measurement apparatus with a repeated accuracy of 0.012 μ m (ST400, NANOVEA, USA).

3. Results and discussion

3.1. Structural characterization of the graphene coated BF

Fig. 2 shows the SEM observation of surface morphology of basalt fibers coating by different concentration of GR. After removing the

commercial coating, Fig. 2a shows the surface of virgin basalt fiber is very smooth. When dipped by polyurethane-only, Fig. 2b shows the topography of basalt fiber become coarse and some sizing polymer can be seen, which indicates the polyurethane sizing agent may have been coated on the surface of basalt fiber. Compared with the virgin and polyurethane-only coating basalt fiber, Fig. 2c–f indicate the surface topography of basalt fiber become much more rough, and the structure of sheets can be observed in the fiber coating apparently. As to the 0.1 GR-BF, a small amount of GR nanosheets can be seen on the surface of BF, and the distribution of GR nanosheets is discontinuous. When the concentration of GR increases to 0.3 g/L and 0.5 g/L, much more GR sheets can be found coating on the surface of basalt fiber, and the entire surface of 0.5 GR-BF is homogeneously covered by GR sheets (Fig. 2e). However, as the concentration of GR increases to 1.0 g/L (Fig. 2f), some GR sheets agglomerates can be observed on the surface of 1.0 GR-BF, which might weaken the interfacial adhesion for the BF reinforced composites to some extent. On the whole, these SEM images reveal that graphene nanosheets can be easily introduced onto the interface regions between basalt fiber and matrix via polyurethane sizing agent, and the interfacial microstructure can be adjusted by varying the content of graphene nanosheets in the sizing agent [35].

Fig. 3a–f shows the SEM observation of cross section of BF coating with different concentration of graphene. According to the SEM images, it can be found that the diameter of virgin basalt fibers, PU and GO modified basalt fibers is 12.5 μ m, 12.9 μ m and 13.8–15.5 μ m, respectively. Estimated from the above results, the thickness of sizing resin on the surface of basalt fibers is about 0.2 μ m, and the thickness of GR is about 0.7–1.5 μ m. Apparently, the content of graphene has a significant influence on the diameter of basalt fiber, and the surface structure of basalt fiber can be easily adjusted by varying the concentration of graphene during the process of modification.

Table 1 shows the surface area of BF, PU-BF and 0.5 GR-BF. The specific surface area of basalt fiber is 9.6 m²/g, while coated with polyurethane and graphene, the specific surface area of PU-BF and GO-BF increases to 11.4 and 18.0 m²/g respectively. Compared with BF, the surface area of 0.5 GR-BF is increased by 87%. This means coating GR via PU can improve the specific surface area of basalt fiber effectively. It will increase the contact area and mechanical meshing effect between the basalt fiber and resin matrix, thus further improve the interfacial bonding between the basalt fiber and resin matrix. As a result, the mechanical, tribological and other performance of basalt fiber reinforced polymer composites can be enhanced.

3.2. Mechanical properties of the BF/PA6 composites

The tensile and flexural tests were carried out to assure the effect of GR on the mechanical properties of GR-BF/PA6 composites. Fig. 4 and Fig. 5 show the results of tensile and flexural properties, respectively. With the increase of GR content, Fig. 4 shows the tensile strength and tensile modulus of GR-BF/PA6 composites increase gradually. Compared with BF/PA6, the tensile strength and tensile modulus of BF/PA6 are increased from 101.8 MPa to 1216.1 MPa to 120.3 MPa and 1506.9 MPa, which has an increase of about 18.2% and 23.9%, respectively. Similarly, the flexural strength and flexural modulus of GR-BF/PA6 also increase gradually from 132.9 MPa to 4002.3 MPa at BF/PA6 composites to 178.2 MPa and 6404.8 at 0.5 GR-BF/PA6 composites, with the increase of about 34% and 60%, respectively. The variation trend of tensile and flexural properties with GR content was coherent with the fiber surface morphology shown in Fig. 2. It is clearly that the coating of GR on the surface of BF contributes to the improvement of tensile and flexural properties for the composites. This demonstrates once again the interfacial properties have an important effect on the mechanical properties of BF/PA6 composites.

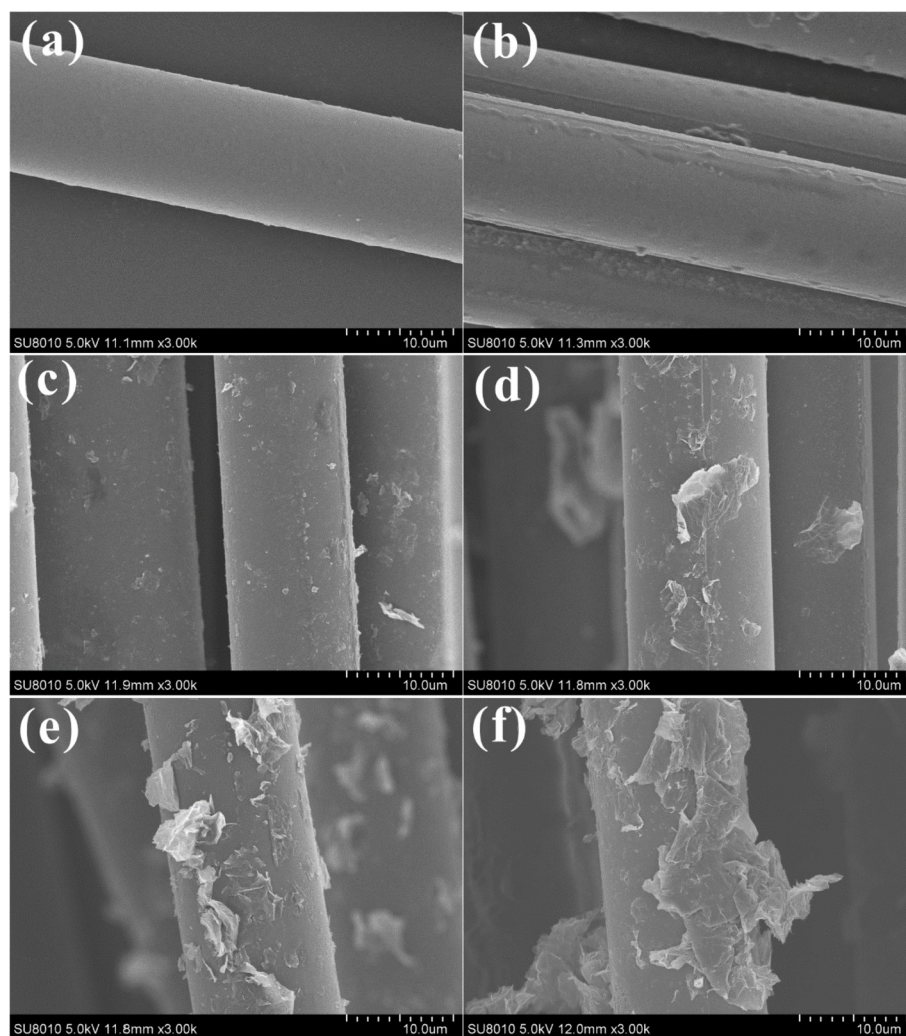


Fig. 2. SEM observation of BF coated with different concentrations of graphene solutions: (a) BF, (b) PU-BF, (c) 0.1 GR-BF, (d) 0.3 GR-BF, (e) 0.5 GR-BF and (f) 1.0 GR-BF.

The DMA results of BF/PA6, PU-BF/PA6, and 0.5 GR-BF/PA6 composites are given in Fig. 6, which exhibit evident changes in dynamic mechanical behavior for the PU modification and GR coating in the interface of composites. Among the three samples, Fig. 6a shows 0.5 GR-BF/PA6 has the highest storage modulus, which is consistent with the results of tensile and flexural testing. As to Fig. 6b, the incorporation of GR onto the surface of BF can significantly enhance the Tg of its resulting composites. Compared to BF/PA6, the Tg of 0.5 GR-BF/PA6 increases from 64.7 to 88.8 °C, an increase of 37.2%. During processing, strong interface can be generated by the interaction between GR and molecular chain segments of PA6 (see Fig. 7), which will confine the vibration of PA6 molecular chains. In addition, the coating of GR in 0.5 GR-BF/PA6 can hinder the mobility of PA6 chains. The coating of GR can enlarge the specific surface area of BF obviously. Generally, the larger the surface area, the higher the Tg. Therefore, the introduction of GR can improve the interface adhesion between BF and PA6 matrix effectively.

To further understand the interface performance and improving mechanism of multiscale graphene-basalt fiber reinforced PA6

composites, we examined their tensile-fractured surfaces by SEM. The SEM micrographs of fracture surfaces are shown in Fig. 7. As evident by the smooth surfaces of BF and debonding phenomenon between BF and PA6 matrix (Fig. 7a and b), it indicates a weak interface, which may be due to the lack of strong bonding between PA6 matrix and untreated BF. In comparison with 0.5 GR-BF/PA6 and 1.0 GR-BF/PA6, the phenomenon of matrix debonding from fiber surface was obviously decreased (Fig. 7e and f). It indicates that the GR sheets can improve the strength and toughness of interfacial region surrounding BF. In this case, the fracture failure occurs within the matrix nearby the surface of BF. So, a highly desirable stronger interfacial bonding between BF and matrix via the introducing of GR may be necessary to improve the mechanical properties for the basalt fiber reinforced composites.

3.3. Tribological properties

The friction coefficients and wear rates versus graphene contents for GR-BF/PA6 composites in dry sliding conditions are plotted in Fig. 8. The incorporation of graphene significantly reduces the friction

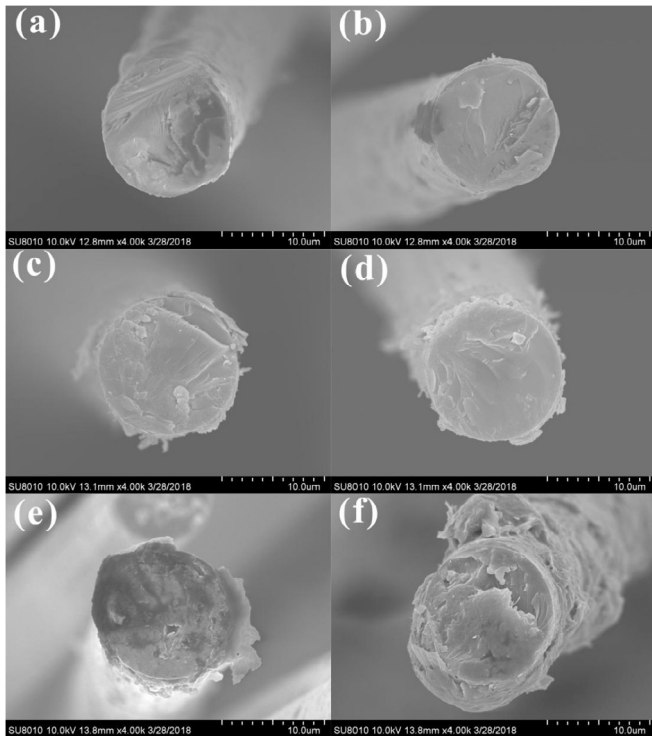


Fig. 3. SEM observation of cross section of BF coated with different concentrations of graphene solutions: (a) BF, (b) PU-BF, (c) 0.1 GR-BF, (d) 0.3 GR-BF, (e) 0.5 GR-BF and (f) 1.0 GR-BF.

Table 1

Specific surface area of BF coated with different concentrations of graphene solutions.

Samples	BF	PU-BF	0.5 GR-BF
Specific surface area (m ² /g)	9.6	11.4	18.0

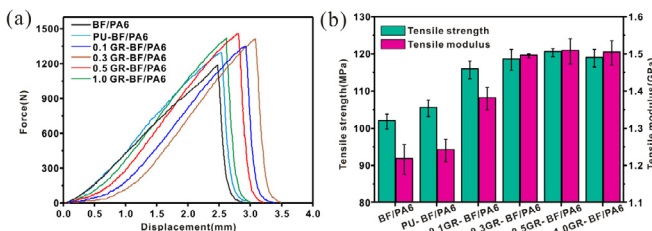


Fig. 4. (a) Tensile stress-strain curve, (b) Tensile strength and tensile modulus of the BF/PA6 composites.

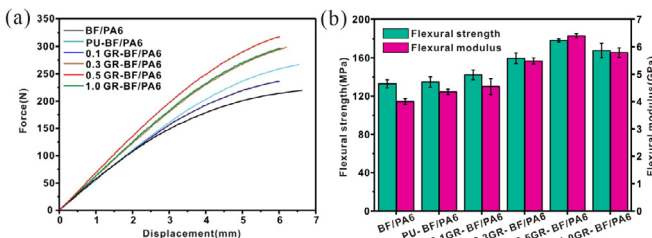


Fig. 5. (a) Flexural stress-strain curve, (b) Flexural strength and flexural modulus of the BF/PA6 composites.

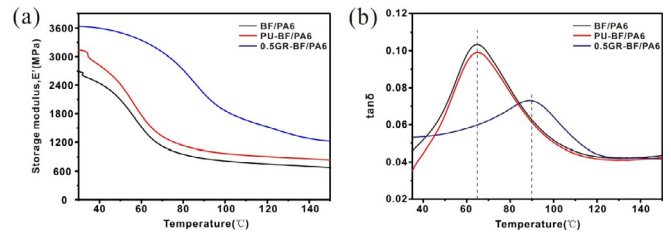


Fig. 6. (a) Storage modulus (E') and (b) $\tan \delta$ vs temperature curves of BF/PA6, PU-BF/PA6, and 0.5 GR-BF/PA6 composites.

coefficient of GR-BF/PA6 composites. As indicated in Fig. 8a, generally, the friction coefficient of GR-BF/PA6 composites decreases with the increment of GR content, and the composite of 0.5 GR-BF/PA6 reaches a fairly low friction coefficient of 0.24, which may be attributed to the lubricating effect of GR and improved interfacial property between BF and PA6 matrix. The friction coefficient slightly increases after the content of GR beyond 0.5 g/L as shown in Fig. 2f. Fig. 8b indicates that the wear rate of GR-BF/PA6 composites also decreases significantly with the increase of GR concentration. Compared with BF/PA6, the wear-resisting property of 0.1 GR-BF/PA6, 0.3 GR-BF/PA6, 0.5 GR-BF/PA6 and 1.0 GR-BF/PA6 are improve by 20.4%, 28.2%, 37.4% and 42.3%, respectively. The mechanical meshing nanoscale interfacial area was generated and bridged by graphene between BF and PA6 matrix, resulting in strong interfacial adhesion and effective stress transfer in the composites. This interfacial structure constructed by graphene between basalt fiber and PA6 matrix can form a stereoscopic structure similar to the gear structure. When the composite is subjected to external force, the external force can be transmitted from the nylon 6 to the fiber through the graphene, similar to the mechanical meshing transmission of gear. Besides, with the highly unfolded feature and flexibility of the GR deformed under the cyclic stress and the improved interfacial adhesive between BF and PA6 matrix, the cracks initiation and growth (Fig. 9e and f) can be limited effectively, which will avoid the brittle feature and relieve the debonding between BF and PA6 matrix. Consequently, the GR-BF/PA6 composites obtained a fairly low friction coefficient and wear rate. Therefore, the coating of GR on the surface of BF plays an important role in enhancing the tribological properties of BF/PA6.

Fig. 9 shows SEM micrographs of worn surface of BF/PA6 and GR-BF/PA6 composites. The worn surface of BF/PA6 presented broadly surface damage and obvious stripping of fibers from the matrix shown in Fig. 9a, with the typical worn form of fatigue wear. It indicated that BF/PA6 had a poor load carrying capacity and its tribological property was not very good. Fig. 9b shows the delaminations of PU-BF/PA6 become relatively less compared with BF/PA6. It also can be seen that many fibers are stripped from the matrix. It may be due to the weak interface bonding strength between polyurethane and PA6 matrix. The wear mechanism of PU-BF/PA6 composites is main of adhesive wear. As shown in Fig. 9c–f, many fibers were embedded on the worn surface and the bonding between BF and PA6 matrix was good. The mechanical meshing nanoscale interfacial area between BF and PA6 matrix bridged by graphene will result in strong interfacial adhesion and effective stress transfer in the composites. And with the improved interfacial adhesive and effect of stress transfer, these fibers can bear much more stress, and the fracture and stripping of fibers and damage of matrix will be relieved. Furthermore, with the good dispersion and self-unfolded feature of GR-BF multi-scale reinforcement in the PA6, the shearing of worn surface will be weakened [36,37]. Therefore, the GR-BF/PA6 behaves lower friction coefficient and better wear resistance.

Fig. 10 shows the worn surface profiles of the composites measured using the laser microscopic 3D and profile measurement apparatus. In

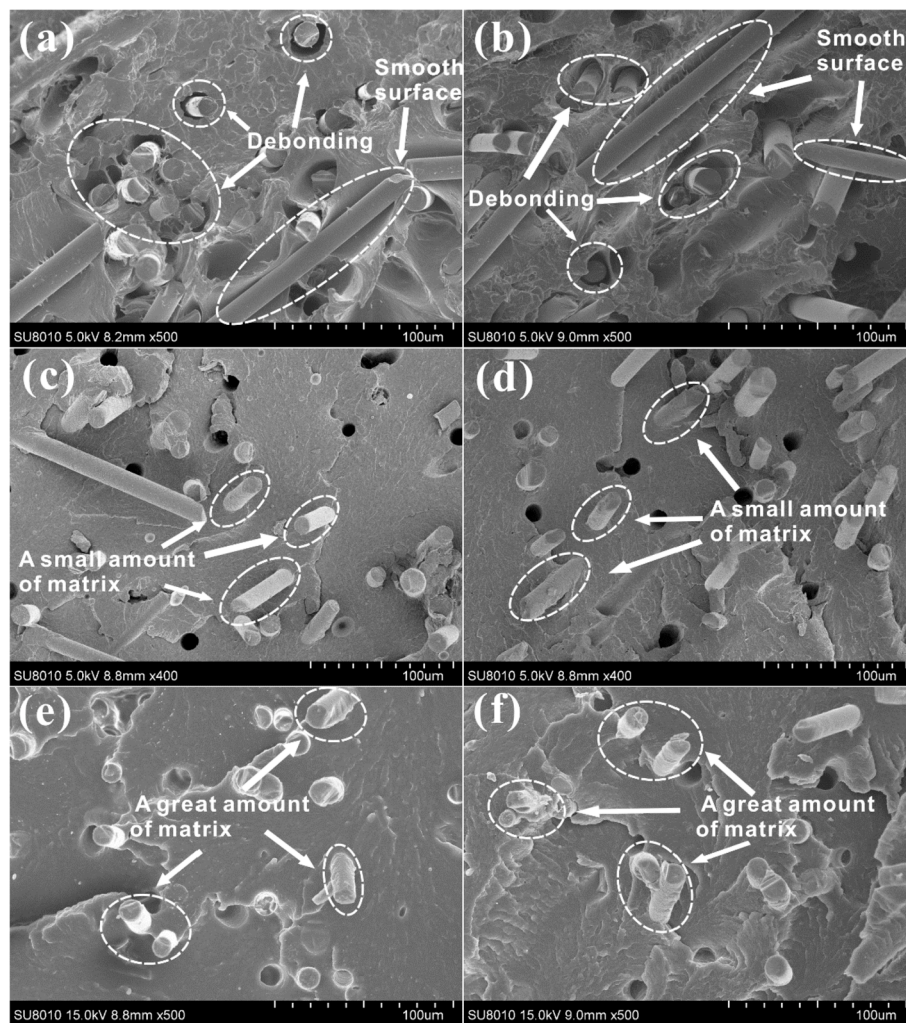


Fig. 7. SEM micrographs of the tensile-fractured surfaces for (a) BF/PA6, (b) PU-BF/PA6, (c) 0.1 GR-BF/PA6, (d) 0.3 GR-BF/PA6, (e) 0.5 GR-BF/PA6 and (f) 1.0 GR-BF/PA6 composites.

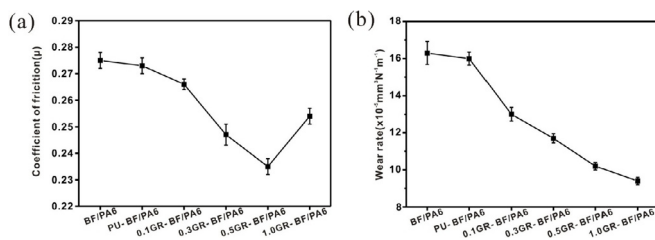


Fig. 8. (a) Friction coefficient and (b) wear rate of the BF/PA6 composites (100 N, 0.167 m/s, 30min).

which, Fig. 10a–c are the worn surface of BF/PA6, PU-BF/PA6 and 0.5 GR-BF/PA6, respectively. The surface roughness parameters (S_a , S_p , S_z , and S_q denote the arithmetic mean height, the maximum peak height, the maximum height and the root mean square height of the wear surface roughness, respectively.) at the location of the horizontal line, which presented in the tables in Fig. 10 shown that the surface

roughness of the PU-BF/PA6 and 0.5 GR-BF/PA6 are lower than BF/PA6 composites. It means that the interfacial modification of PU or GR can improve the wear resistance of BF/PA6. In comparison, the surface roughness of 0.5 GR-BF/PA6 displays a much lower value, which may be reasoned by the better interfacial adhesive between BF and PA6 matrix modified by GR and the excellent lubrication of graphene on the worn surface, thus the graphene coated BF can reduce the surface roughness of wear surface effectively [38,39].

4. Conclusions

Multiscale graphene-basalt fiber reinforcement was prepared by a facile dipping method via polyurethane sizing agent. The graphene on the surface of basalt fibers was homogeneous, and the microstructure of the interfacial region between basalt fibers and matrix could be adjusted by varying the content of graphene nanosheets in the sizing agent. The effects of multiscale graphene-basalt fiber reinforcement on mechanical and tribological properties of PA6 composites were studied by tensile and flexural tests, DMA test, and friction and wear tests. Owing to the bridge effect of graphene, a mechanical meshing

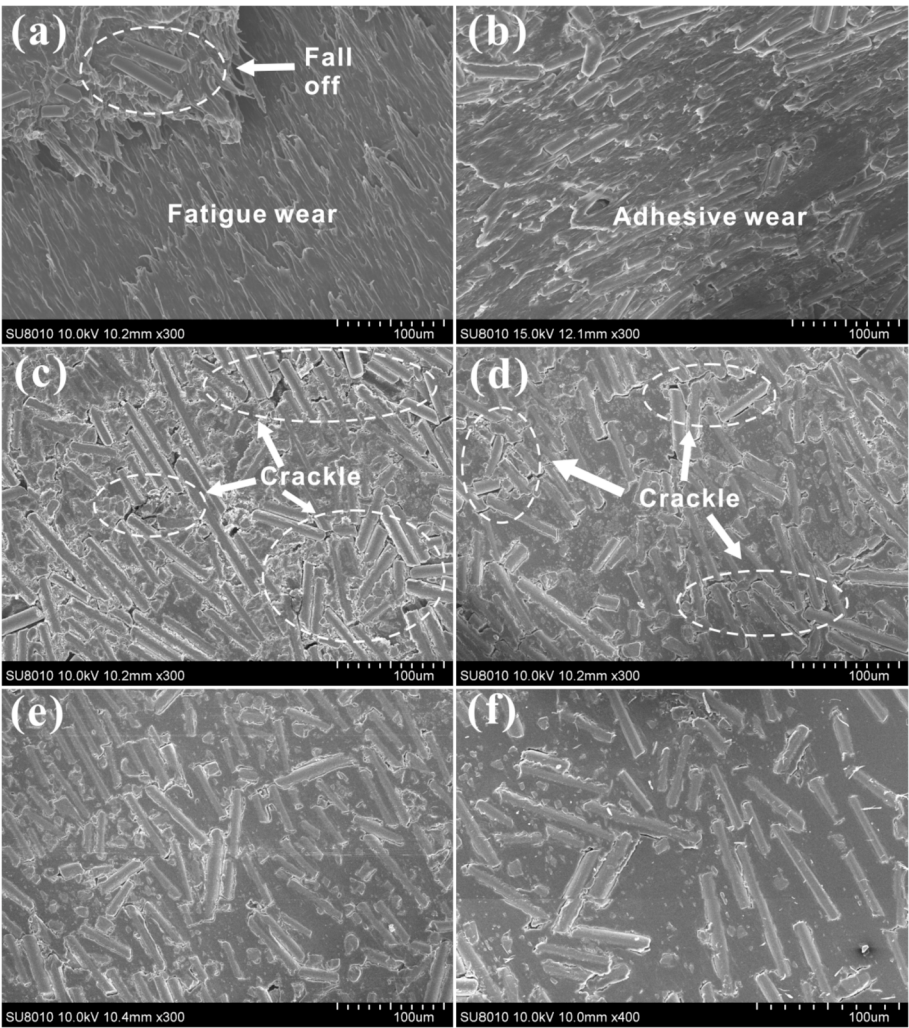


Fig. 9. SEM micrographs of worn surfaces of (a) BF/PA6, (b) PU-BF/PA6, (c) 0.1 GR-BF/PA6, (d) 0.3 GR-BF/PA6, (e) 0.5 GR-BF/PA6 and (f) 1.0 GR-BF/PA6 composites.

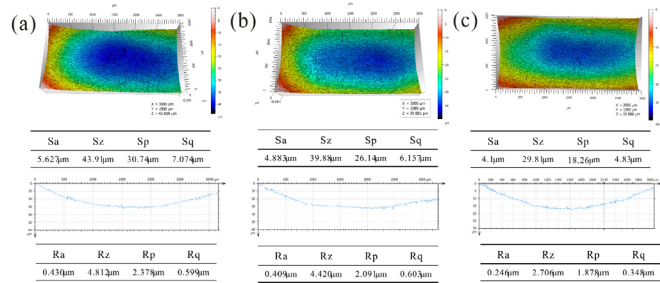


Fig. 10. 3D topographies of worn surface: (a) BF/PA6, (b) PU-BF/PA6 and (c) 0.5 GR-BF/PA6.

nanoscale interfacial area was generated between BF and PA6 matrix, which may enhance the interfacial adhesive between basalt fibers and PA6 matrix, and the stress transfer of the corresponding composites could be improved. The uniform coating graphene on the surface of basalt fibers led to the increase of storage modulus and Tg for the BF/PA6 composites and 18.2% and 34% improvement for their tensile and flexural strength. And the coating of GR on the surface of BF also played an important role in enhancing the tribological properties of BF/PA6. Compared with BF/PA6, the GR-BF/PA6 composites obtained a fairly low friction coefficient and highly wear-resisting property, which may be reasoned by the improved interfacial adhesion, effective stress transfer and the highly unfolded feature and flexibility of the graphene deformed under the cyclic stress during the friction process. Therefore, this work presented a facile approach for enhancing the mechanical and tribological properties of basalt fiber-reinforced composites by introducing graphene into their interface to form a multiscale structure.

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